

Data Science Capstone Project Report

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Project Repository:

<https://github.com/media4everyone/Data-Science-capstone-project-report>

Section 1: Executive Summary

This Data Science Capstone Project aims to predict the success of SpaceX Falcon 9 first-stage landings to determine the reusability and commercial cost per launch. By leveraging data collection via SpaceX REST API and Wikipedia Web Scraping, followed by comprehensive Data Wrangling, Exploratory Data Analysis (EDA) using SQL and Visualization tools, Interactive Visual Analytics with Folium and Plotly Dash, and Predictive Classification Modeling, we achieved a high-accuracy predictive model.

Key Results & Methodology Highlights:

- **Data Collection:** Collected 94 Falcon 9 launch records via SpaceX REST API and 121 records via Web Scraping (BeautifulSoup).
- **Data Wrangling:** Handled missing values (imputed Payload Mass with mean values) and constructed the target binary variable Class (1 for successful first-stage landing, 0 for failure/unsuccessful).
- **Exploratory Data Analysis:** Uncovered significant correlations between Payload Mass, Orbit type (e.g., ES-L1, GEO, HEO, SSO achieved near 100% success rate), and Launch Sites (KSC LC-39A leading in success rates).
- **Interactive Maps & Dashboard:** Built geospatial visualizations using Folium to evaluate proximity to coastlines and transportation hubs, and created interactive Plotly Dash applications for real-time payload filtering.
- **Predictive Analytics:** Built and evaluated Logistic Regression, Support Vector Machine (SVM), Decision Tree, and K-Nearest Neighbors (KNN) classifiers tuned with GridSearchCV. All models achieved a top validation accuracy of **83.33%**. Decision Tree was selected as the optimal model due to high interpretability and minimal execution latency.

Section 2: Introduction & Problem Statement

Background

SpaceX has revolutionized the space transport industry by significantly reducing space launch costs. A major driver behind these savings is the reusability of the Falcon 9 rocket's first stage. While conventional rocket stages are discarded after launch, SpaceX successfully lands its first-stage boosters back on land or on autonomous spaceport drone ships (ASDS), allowing them to be refurbished and flown again.

Problem Statement

While SpaceX advertises Falcon 9 launches at an industry-disrupting price of \$62 million (compared to \$165+ million from traditional providers), a critical factor in determining total launch cost is whether the first stage will be successfully recovered. If we can accurately predict whether a Falcon 9 first stage will land successfully, we can estimate the actual cost of a launch. This insight allows competing companies or commercial clients to make data-driven bidding and operational decisions.

Section 3: Data Collection – SpaceX REST API

We programmatically retrieved historical launch data using the SpaceX REST API endpoints (<https://api.spacexdata.com/v4/launches/past>).

API Data Collection Workflow / Pipeline

1. **API Request:** Send HTTP GET requests to the SpaceX v4 API endpoint.
2. **JSON Parsing:** Parse nested JSON payloads containing launch parameters, rocket details, payload information, launchpads, and core landing outcomes.
3. **Data Extraction:** Extract key features including FlightNumber, Date, BoosterVersion, PayloadMass, Orbit, LaunchSite, Outcome, Flights, GridFins, Reused, Legs, LandingPad, Block, ReusedCount, Serial, Longitude, and Latitude.
4. **Filtering:** Filter out Falcon 1 launches, retaining exclusively Falcon 9 records for analysis.

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Section 4: Data Collection – Web Scraping

To complement API data and ensure historical completeness, we scraped Falcon 9 launch records from Wikipedia using BeautifulSoup4 and requests.

Web Scraping Process

- **HTML Fetching:** Downloaded HTML pages covering Falcon 9 and Falcon Heavy launch history.
- **DOM Parsing:** Created a BeautifulSoup object to parse HTML table structures (class="wikitable").
- **Data Extraction:** Extracted table column headers and iterated over rows to capture Flight Number, Date & Time, Booster Version, Launch Site, Payload, Payload Mass, Orbit, Customer, and Landing Outcome.
- **Data Standardisation:** Cleaned scraped text strings, removed reference tags (e.g., [1]), parsed numerical payload values, and formatted dates into standardized ISO formats.

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Section 5: Data Wrangling Methodology

Data quality and consistency were established through rigorous cleaning and feature engineering in Pandas.

Cleaning & Feature Engineering Steps

- **Handling Missing Values:** Identified missing numerical entries in PayloadMass and imputed them using the mean payload mass calculated across the dataset.
- **Target Encoding (Class Generation):** Converted categorical Landing Outcome into a binary target variable Class:
 - Class = 1: Successful landing (e.g., True Ocean, True RTLS, True ASDS).
 - Class = 0: Unsuccessful or no landing attempt (e.g., False Ocean, False RTLS, False ASDS, None ASDS, None None).
- **One-Hot Encoding:** Applied pd.get_dummies() to categorical features (Orbit, LaunchSite, LandingPad, Serial) to prepare numerical inputs for machine learning algorithms.
- **Standardization:** Standardized numerical features using StandardScaler prior to fitting classification models.

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Section 6: Exploratory Data Analysis (EDA) with Data Visualization

Visual exploration was conducted using Seaborn and Matplotlib to identify patterns across flight numbers, payload weights, launch sites, and orbit types.

Key Visualizations & Findings

- **Flight Number vs. Launch Site:** As flight numbers progressed over time, overall landing success rates improved significantly across all launch sites. Early flights at CCAFS SLC-40 showed lower success rates, whereas later flights at KSC LC-39A and VAFB SLC-4E demonstrated high success consistency.
- **Payload Mass vs. Launch Site:** Heavy payloads (>10,000 kg) were primarily launched from KSC LC-39A and CCAFS SLC-40, with higher payload weights positively correlating with drone ship (ASDS) landing attempts.
- **Success Rate vs. Orbit Type:** Orbits such as ES-L1, GEO, HEO, and SSO exhibited 100% success rates, while orbits like GTO had moderate success rates due to higher re-entry velocities.
- **Yearly Success Trend:** Line charts tracking landing success over years (2010–2020) demonstrated a clear positive trend, starting at 0% in 2010 and reaching over 80% after 2017.

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Section 7: Exploratory Data Analysis (EDA) with SQL

Structured queries were executed on the SQLite database (SPACEXTABLE) to answer quantitative business questions.

SQL Query Summary

Query Description	SQL Code / Logic Executed	Result Summary
Unique Launch Sites	SELECT DISTINCT "Launch_Site" FROM SPACEXTABLE;	CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, CCAFS SLC-40
Launches starting with 'CCA'	SELECT * FROM SPACEXTABLE WHERE "Launch_Site" LIKE 'CCA%' LIMIT 5;	First 5 records from Cape Canaveral sites retrieved successfully
Total Payload Carried by NASA (CRS)	SELECT SUM("PAYLOAD_MASS__ KG_") FROM SPACEXTABLE WHERE "Customer" = 'NASA (CRS)';	45,596 kg total payload mass delivered
Average Payload for Booster F9 v1.1	SELECT AVG("PAYLOAD_MASS__K G_") FROM SPACEXTABLE WHERE "Booster_Version" = 'F9 v1.1';	2,928.4 kg average payload mass
First Successful Ground Landing Date	SELECT MIN("Date") FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Success (ground pad)';	2015-12-22
Successful Drone Ship Boosters (4,000–6,000 kg)	SELECT "Booster_Version" FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Success (drone ship)' AND	F9 FT B1022, F9 FT B1026, F9 FT B1021.2, F9 FT B1031.2

Query Description	SQL Code / Logic Executed	Result Summary
	"PAYLOAD_MASS__KG_" BETWEEN 4000 AND 6000;	
Total Successful vs Unsuccessful Landings	SELECT "Landing_Outcome", COUNT(*) FROM SPACEXTABLE GROUP BY "Landing_Outcome";	Categorized counts of successes, drone ship failures, and ocean landings
Maximum Payload Carrying Boosters	SELECT "Booster_Version" FROM SPACEXTABLE WHERE "PAYLOAD_MASS__KG_" = (SELECT MAX("PAYLOAD_MASS__KG_") FROM SPACEXTABLE);	F9 B5 B1048.4, B1049.4, B1051.3, B1056.4, etc.
2015 Drone Ship Landing Outcomes	SELECT strftime('%m', Date), Launch_Site, Booster_Version FROM SPACEXTABLE WHERE Landing_Outcome = 'Failure (drone ship)' AND strftime('%Y', Date) = '2015';	Identified January and April 2015 failure events
Rank Landing Outcomes (2010-06-04 to 2017-03-20)	SELECT "Landing_Outcome", COUNT(*) AS Count FROM SPACEXTABLE WHERE "Date" BETWEEN '2010-06-04' AND	No attempt (10), Success (drone ship) (5), Failure (drone ship) (5)

Query Description	SQL Code / Logic Executed	Result Summary
	'2017-03-20' GROUP BY "Landing_Outcome" ORDER BY Count DESC;	

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Section 8: Interactive Visual Analytics – Folium Maps & Plotly Dash

Geospatial Analytics with Folium

- **Launch Site Markers:** Plotted interactive map markers for all launch complexes (CCAFS LC-40, KSC LC-39A, VAFB SLC-4E) with color-coded success indicators.
- **Proximity Analysis:** Calculated distance vectors using the Haversine formula from launchpads to nearest coastlines, railways, highways, and cities.
 - *Insight:* All launch sites are located within close proximity (<2 km) to coastlines for safety in case of aborts, and adjacent to railways and highways for logistics transport.

Interactive Dashboard with Plotly Dash

- **Interactive Dropdown:** Select All Sites or specific launch sites to dynamically update visualizations.
- **Success Pie Chart:** Rendered proportions of successful vs. failed launches for selected sites.
- **Payload Range Slider & Scatter Plot:** Rendered interactive scatter plot plotting Payload Mass vs. Landing Class, color-coded by Booster Version category (e.g., v1.0, v1.1, FT, B5).

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Section 9: Predictive Analytics & Machine Learning

Models

We built, hyperparameter-tuned, and evaluated four supervised classification algorithms to predict landing success (Class = 0 or 1).

Model Training & Hyperparameter Tuning

- **Train/Test Split:** Split features (X) and target (Y) into 80% training data and 20% test data using `train_test_split(random_state=2)`.
- **Hyperparameter Tuning:** Executed 10-fold cross-validation using `GridSearchCV` across all model parameter grids.

Model Performance Comparison

Model Algorithm	Best CV Accuracy Score	Test Set Accuracy Score	Key Hyperparameters
Logistic Regression	83.33%	83.33%	C: 0.01, penalty: l2, solver: lbfgs
Support Vector Machine (SVM)	83.33%	83.33%	C: 1.0, gamma: 0.0316, kernel: sigmoid
Decision Tree Classifier	83.33%	83.33%	criterion: gini, max_depth: 4, splitter: best
K-Nearest Neighbors (KNN)	83.33%	83.33%	algorithm: auto, n_neighbors: 10, p: 1

Confusion Matrix Evaluation

All four models produced identical confusion matrices on the test dataset (18 total test samples):

Predicted:	Class 0	Class 1
Actual Class 0:	3	3

Actual Class 1: 0 12

- **True Negatives (TN):** 3 (Unsuccessful landings correctly predicted)
- **False Positives (FP):** 3 (Unsuccessful landings misclassified as successful)
- **False Negatives (FN):** 0 (100% Recall for successful landings)
- **True Positives (TP):** 12 (Successful landings correctly predicted)

Section 10: Conclusion & Innovative Insights

Conclusion & Best Model Selection

All models achieved identical cross-validation and test set accuracies of **83.33%**. However, the **Decision Tree Classifier** is selected as the recommended model for deployment because:

1. It provides clear, interpretable decision rules (node splits based on Payload Mass and Booster Version) that non-technical stakeholders can inspect.
2. It exhibits zero prediction latency and low computational resource requirements during inference.

Key Business & Innovative Insights

- **Heavy Payloads are Safer to Recover:** Payloads in the 3,000 kg – 8,000 kg range showed higher landing success rates on ASDS drone ships compared to light payloads (<2,000 kg).
- **Block 5 Superiority:** The transition to Falcon 9 Block 5 boosters boosted landing reliability to near 100%, proving that hardware iterations successfully solved early re-entry thermal and control issues.
- **Cost Optimization Impact:** With a reliable landing success prediction mechanism (>83% accuracy), launch insurers and commercial launch brokers can dynamically price contracts and save up to \$40 million per reused flight stage.